The Rope Hypothesis
Scientific Glossary — Version 4
Definitions, Physical Identifications, and Cross-References
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HOW TO USE THIS GLOSSARY

This glossary defines 137 terms used in the Rope Hypothesis programme across 16 thematic categories. Terms marked ★ are those where the rope model provides a genuinely new physical explanation not available in the standard account. Terms marked ★NEW were added in v3; the v4 addendum adds the quantum-measurement-arc terms. Terms marked ▲UPDATED have expanded or corrected definitions from v2.

The rope model identifies physical reality with a network of two-strand helical ropes connecting every pair of atoms. The 16 categories below cover: the rope's geometric structure (Cat. 1), waves and particles (Cat. 2–3), spacetime (Cat. 4), forces (Cat. 5–6), relativity (Cat. 7–8), black holes (Cat. 9), chemistry (Cat. 10), thermodynamics (Cat. 11), cosmology and gravity (Cat. 12), quantum mechanics (Cat. 13), nuclear physics (Cat. 14), rope bundle theory (Cat. 15), and the quantum rope field (Cat. 16).

Standard model contrasts (in grey) show how the rope model re-identifies concepts that in standard physics are taken as primitive or unexplained.

1. Foundational Concepts

ROPE

The fundamental physical object. A two-strand helical structure of indefinite length connecting every pair of atoms in the universe. Carries all physical interactions as wave disturbances (kinks, tension gradients, or bundle forces).

Standard model: No analogue. In standard physics, forces are mediated by gauge bosons or field quanta; there is no mechanical medium.

ROPE NETWORK

The totality of all ropes. Every atom is connected to every other atom by a rope. The network is the physical substrate of space, time, and all forces. Its large-scale geometry is equivalent to spacetime; its local properties determine all physical laws.

See also: SPACE, SPACETIME

STRAND

One of the two filaments making up a rope. The two strands, wound helically at winding angle θ_w, give the rope its two degrees of freedom. In the quantum theory: strand 1 → right-mover Weyl spinor χ_R; strand 2 → left-mover χ_L. The two-component structure is responsible for fermionic statistics and parity violation.

See also: WINDING ANGLE, WEYL SPINOR, CHIRALITY

TENSION T [N]

The elastic restoring force per unit length along the rope, equal to energy per unit length [J/m = N]. Sets the wave speed $c = \sqrt{T/\mu}$. Determines all forces at the relevant scale via $T = u \times \lambda_c^2$.

$T = u \times \lambda_c^2$ [energy density × coherence area]

See also: $T = u \times \lambda_c^2$, ENERGY DENSITY, COHERENCE LENGTH

LINEAR DENSITY μ [kg/m]

Mass per unit length of the rope. Together with tension T, determines wave speed $c = \sqrt{T/\mu}$. The ratio $T/\mu = c^2$ is fixed by the speed of light. In terms of energy density: $\mu = u\,\lambda_c^2/c^2$.

WINDING ANGLE θ_w

The angle between each strand and the rope's long axis, measuring the tightness of the helix. A fundamental geometric parameter of the rope. Appears in: cosmic birefringence (Chern-Simons coupling $\propto \sin(2\theta_w)$), electroweak mixing (Weinberg angle $\theta_W = \theta_w$), and parity violation (left-handed helix). Two independent measurements give $\theta_w \approx 28.7°–30°$.
See also: WEINBERG ANGLE, COSMIC BIREFRINGENCE, CHIRALITY

HELIX RADIUS r_h [m]
The distance between the rope axis and each strand. Constrained by Coulomb precision tests to $r_h < 10^{-18}$ m. Together with rope density n_rope, determines the birefringence amplitude: $\beta \propto n_{rope}\, r_h^2 \sin(2\theta_w)$.

CHIRALITY
The handedness of the rope helix. The rope is left-handed (by convention). Physical consequence: the weak force couples only to left-strand excitations (χ_L), producing the Standard Model's V−A parity violation. A right-handed helix would couple to χ_R — but the universe has a specific chirality.
See also: PARITY VIOLATION, WEAK FORCE

INTERPENETRABILITY / IMPENETRABILITY ▲UPDATED
In Gaede's original formulation: individual ropes interpenetrate freely (unlike matter). This is correct for single ropes. HOWEVER: the rope bundle theory shows that dense collections of ropes ($N_{eff} \gg 1$) generate CONTACT FORCES that resist penetration. Macroscopic solidity, hardness, and impenetrability emerge from the contact condition $N_{eff} = n_{rope} \times \lambda_c^3 \gg 1$. Individual ropes interpenetrate; rope bundles resist penetration when the coherence volume contains many ropes.
See also: CONTACT CONDITION, N_EFF, ROPE BUNDLE

COHERENCE LENGTH λ_c [m] **★NEW**
The length scale at which rope waves contribute coherently to bundle contact forces. For a particle of mass m: $\lambda_c = \hbar/(mc)$ (Compton wavelength). Examples: pion Compton $\lambda_\pi = 1.46$ fm (nuclear scale); $\alpha\, a_0 = 3.9\times10^{-13}$ m (atomic scale). Sets the range of forces at each scale.
See also: CONTACT CONDITION, $T = u\times\lambda_c^2$

COHERENCE AREA λ_c^2 [m²] **★NEW**
The cross-sectional area associated with one rope at the contact threshold. The bridge between rope tension T [J/m] and energy density u [J/m³]: $T = u \times \lambda_c^2$. At nuclear scale: $\lambda_c^2 = (1.46\ \text{fm})^2 = 2.1\times10^{-30}$ m². At atomic: $(3.9\times10^{-13}\ \text{m})^2 = 1.5\times10^{-25}$ m².

CONTACT CONDITION $N_{eff} \gg 1$ **★NEW**
The criterion for a rope bundle to be in the dense (classical contact force) regime: $N_{eff} = n_{rope} \times \lambda_c^3 \gg 1$. Below this, residual (quantum) forces apply. Nuclear matter at saturation density has $N_{eff} \approx 0.4–1.8$, just at the contact threshold for pion-scale coherence. This explains why nuclear matter saturates and why the nuclear force has finite range.
See also: ROPE DENSITY, N_EFF, TWO-LEVEL FORCE STRUCTURE

ENERGY DENSITY u [J/m³] **★NEW**
Energy per unit volume at a given physical scale. Determines rope tension via $T = u \times \lambda_c^2$. Examples: $u_{nucleon} = m_p c^2/V_{proton} = 5.5\times10^{34}$ J/m³; $u_H = 13.6\ \text{eV}/(4\pi a_0^3/3) = 3.5\times10^{12}$ J/m³; $u_{cosmic} = \Omega_m \rho_{crit} c^2 = 2.4\times10^{-10}$ J/m³.

$T = u \times \lambda_c^2$ ★NEW
The central relation connecting rope tension to physical energy scales. Derived from $T = u/n_{area}$ with $n_{area} = 1/\lambda_c^2$ at the contact threshold. Confirmed at three scales: nuclear ($T = 1.17\times10^5$ N $\approx k_{QCD} = 1.46\times10^5$ N, within 20%); atomic (T_{EM} recovers $I_H = 13.6$ eV

exactly, 0.00 ppm); cosmological ($T_{cosmic} = 6.1\times10^{-18}$ N, sets birefringence energy scale). Spans 23 orders of magnitude in T.

T_nuclear = u_nucleon × (hbar/m_pi c)^2 = 1.17e5 N [k_QCD within 20%]

T_EM = u_H × (alpha*a0)^2 = 5.2e-13 N [I_H exact]

2. Light and Waves

ROPE BUNDLE

The collection of ropes running between two atomic hubs. Every pair of atoms is connected by one rope, but from the perspective of any one atom, its hub is the convergence point of ropes going to every other atom. Between any two nearby atoms, the ropes collectively form a bundle. When the effective rope count $N_{eff} = n_{rope} \times \lambda_c^3 \gg 1$, the bundle is in the dense regime and generates Level 1 contact forces (solidity, QCD string tension). Below this threshold, forces are quantum residual. The rope bundle is the physical basis for what standard physics calls the electromagnetic field, strong force, and gravitational field between two atoms.

ROPE WAVE

A transverse displacement wave propagating along the rope at speed $c = \sqrt{T/\mu}$. Two polarisations (x and y displacements), corresponding to the two Pauli matrix directions. Electromagnetic waves are transverse rope waves. Sound is not a rope wave but a pressure wave in rope bundles.

PILOT WAVE

The rope wave that guides the motion of the kink (particle). In the two-object rope picture: the rope wave (extended, wavelike) is the pilot wave; the kink (localised, particle-like) is guided by it. This is the Bohmian interpretation of quantum mechanics, with the rope as the guiding medium.

See also: PHOTON, BORN RULE, WAVEFUNCTION

POLARISATION

The direction of the rope's transverse displacement. Two independent polarisations (linear x and y) combine to give circular polarisation. In the Dirac/Weyl picture: the two polarisations correspond to the two strand displacements u_1 (x) and u_2 (y). The helical coupling between strands rotates linear into circular polarisation.

WAVE SPEED c [m/s]

The speed of propagation of rope kinks and waves: $c = \sqrt{T/\mu}$. Universal because $T/\mu = c^2$ is the same for all ropes in the network. Finite because the rope has finite tension and inertia.

Standard model: c is a postulated constant.

FREQUENCY $\nu = \omega/(2\pi)$ [Hz]

The oscillation rate of the rope wave. For a photon: $E = h\nu = \hbar\omega$. In rope language: the rate at which the kink passes through one wavelength of its pilot wave.

WAVELENGTH λ [m]

The spatial period of the rope wave: $\lambda = c/\nu$. De Broglie wavelength of a massive particle: $\lambda = h/(mv)$. In rope language: the distance between successive identical phases of the pilot wave guiding the kink.

PHOTON

A massless transverse kink propagating along a rope at speed c. DISTINCT FROM ATOMS: a photon is a wave disturbance on a rope; an atom is a permanent topological knot (hub) in the rope network. A photon travels between two atomic hubs along the rope connecting them. It is created when an atomic hub is disturbed (emitting a kink) and absorbed when another hub is disturbed (absorbing the kink). Spin-1: it takes one full rope oscillation to return to the original

state, giving exchange phase +1 (bosonic). The two polarisation states are the two transverse displacement directions (x and y). The photon does not exist between emission and absorption as a persistent object — it is a propagating pattern in the rope between the two atomic hubs.
Standard model: A massless spin-1 gauge boson of the electromagnetic field.
See also: KINK, PILOT WAVE, POLARISATION, SPIN

3. Matter and Particles

PROTON STAR / CONVERGENCE ★NEWv4

An atom's core pictured as Gaede draws it: a convergence of very many ropes onto a focal knot inside a woven shell, with ropes radiating outward — a many-rope "star," not a small few-rope tangle. The atomic size is the scale of this convergence pattern (a standing-wave wavelength), NOT the spacing of the ropes themselves, which is far finer (sub-nuclear). Atoms are thus coarse structures in a much finer rope mesh.
See also: PACKING COUNT, STANDING-WAVE SHELL, SUB-NUCLEAR MESH

PACKING COUNT N ★NEWv4

The number of ropes converging into an atomic core, distinct from the charge/winding number. A geometric packing argument gives the pattern radius $R \approx a\cdot\sqrt{N/4\pi}$, so atomic size grows as the square root of the rope count. Consistent with the Bohr radius for $N \sim 10^{12}$ ropes at mesh scale $a \sim 10^{-16}$ m — a plausible model, not a derivation: N is not yet fixed by independent physics (notably, it is NOT the charge, which is the winding number = 1 for hydrogen).
See also: PROTON STAR / CONVERGENCE, SUB-NUCLEAR MESH, WINDING NUMBER

QUARK SUB-KNOT ★NEWv4

A quark identified as a partial rope knot — a sub-knot carrying fractional winding (+2/3, −1/3) that is not stable alone and only settles when it completes a whole-integer knot with its partners. Confinement follows: a piece of a knot cannot stand by itself. Fractional windings are consistent with integer charge quantisation because they are always confined and sum to integers (proton uud = +1, neutron udd = 0), but they require a more general topological classification than integer linking — an extension not yet built.
Standard model: quarks are elementary fields with fractional charge; confinement is observed and reproduced numerically but has no simple mechanical statement.
See also: CONFINEMENT, NUCLEON KNOT, CHARGE AS HANDEDNESS / LINKING

PROTON

A topological rope knot at an atomic hub, with net winding number $n = +1$, composed of three quark sub-knots (u, u, d with winding numbers +2/3, +2/3, −1/3). The quark sub-knots are connected by QCD string rope bundles internal to the hub. The proton is the simplest stable hub structure with $n = +1$. Proton mass 938.3 MeV is the elastic energy of the three-quark knot system — the energy stored in the topological distortion of the ropes at the hub. The proton is NOT a kink travelling on a rope; it is a permanent topological structure at a network junction.
See also: QUARK, QCD STRING, TOPOLOGICAL CHARGE

MASS m [kg]

The resistance of a topological rope knot (atom, nucleus, or electron) to acceleration, arising from the coupling of left-mover and right-mover rope excitations within the knot structure. In the Dirac rope picture: mass = inter-strand coupling frequency $m = \hbar\Omega\_coupling/c^2$. For massless photons: no inter-strand coupling, no mass. For massive particles: the background rope network couples the two strands of the knot, giving it inertia. Mass is the energy of the topological distortion at the hub divided by c^2: $m = E\_knot/c^2$.

Standard model: Mass is a primitive quantity in Newtonian mechanics. The Higgs mechanism explains it in the Standard Model as a coupling to the Higgs field.
See also: HIGGS MECHANISM, DIRAC EQUATION, ZITTERBEWEGUNG
REST ENERGY $E = mc^2$
The elastic energy stored in the rope knot that constitutes the particle. $E = mc^2$ is derived from the work-energy theorem applied to the rope: accelerating a kink increases the tension gradient, which does work. For matter-antimatter annihilation: the rope knot ($n = -1$) meets its antiknot ($n = +1$), they cancel ($n = 0$), and the stored elastic energy releases as two photon kinks.
CHARGE e [C] **▲UPDATED**
The topological winding number of the rope knot, quantised because winding numbers are integers or fractions. Electron: $n = -1$ (charge $-1e$). Proton: $n = +1$. Up quark: $n = +2/3$. Down quark: $n = -1/3$. Conservation of charge = conservation of total winding number at any interaction vertex. Confirmed in beta decay: $d \to u + W^-$ gives $(-1/3) = (+2/3) + (-1)$.
See also: TOPOLOGICAL CHARGE, BETA DECAY
SPIN ▲UPDATED
The intrinsic angular momentum from the two-strand helix geometry. Spin-1/2 (electron): half-twist of the rope helix during exchange → exchange phase $e^{i\pi} = -1$ → fermionic statistics (Pauli exclusion). Spin-1 (photon): full twist → phase $e^{2i\pi} = +1$ → bosonic. Spin-0 (pion): no twist → bosonic. The spin-statistics theorem is a consequence of rope topology: half-integer spin from half-twists, integer from integer twists.
See also: FERMIONIC STATISTICS, PAULI EXCLUSION, TWO-STRAND ROPE
ANTIMATTER ▲UPDATED
Particles with the opposite topological winding number. Positron (antielectron): $n = +1$. Antiproton: $n = -1$. In the Dirac equation: the negative-energy solutions correspond to the $n > 0$ winding sector. Pair annihilation: $(-1) + (+1) = 0$, knots cancel, elastic energy releases as two photon kinks. Pair creation: a photon kink with $E \geq 2m_e c^2$ creates a knot-antiknot pair from rope elastic energy. Antimatter is not mirror matter — it is the topological complement.
See also: WINDING NUMBER, PAIR CREATION/ANNIHILATION
PAIR CREATION/ANNIHILATION ★ ★NEW
Pair annihilation: two rope knots with opposite winding numbers meet and cancel topologically. The rope straightens; elastic energy stored in both knots releases as two photon kinks. Total winding conserved: $(-1) + (+1) = 0$. Pair creation (reverse): a photon kink $E \geq 2m_e c^2$ distorts the rope sufficiently to create a knot-antiknot pair, storing the photon energy as rope elastic energy. $E = 2m_e c^2 = 1.022$ MeV is the minimum threshold.
See also: ANTIMATTER, REST ENERGY, TOPOLOGICAL CHARGE
ATOM
The hub of the rope network: the point at which an enormous number of ropes converge, one rope connecting this atom to every other atom in the universe. The atom's identity is the topological character of this hub — the winding numbers and knot structure of the ropes at the junction. The NUCLEUS is a multi-quark knot assembly at the hub centre, held together by QCD string bundles (rope bundles between the quark sub-knots). The ELECTRONS are rope kinks — transverse topological distortions — guided by standing-wave rope modes (orbitals) around the nuclear hub. Moving an atom means moving the hub: all ropes connected to it adjust their geometry to the new position. This is not elastic stretching but a geometric rearrangement of the network around the moved junction. IMPORTANT DISTINCTION: the atom is the

hub/knot, NOT a kink. Photon kinks travel along the ropes between hubs. Atoms are the hubs themselves.
See also: ATOMIC ORBITAL, COVALENT BOND, QCD STRING
KNOT ★
The topological structure at an atomic hub that gives the atom its physical identity. A knot is a permanent distortion of the rope's winding topology — it cannot be removed by continuous deformation without breaking the rope. The electron is a knot with winding number $n = -1$; the positron $n = +1$; the proton is a three-sub-knot assembly with net $n = +1$. KNOT vs KINK: a kink is a transient wave disturbance (photon, propagating at c); a knot is a permanent topological feature (particle, at rest or $v < c$). A kink can pass through a point and leave; a knot cannot be removed without pair annihilation. The knot stores the particle's rest energy as rope elastic energy. When a particle moves, the knot changes position in the network; when a photon moves, the kink propagates along a rope between two knot-hubs. This is the fundamental distinction between matter and radiation in rope theory.
See also: ATOM, PHOTON, CHARGE, TOPOLOGICAL CHARGE, PAIR CREATION/ANNIHILATION
4. Space and Time
SPACE
The set of all rope connections between atoms. Points in space are intersections of rope paths; distances are rope lengths. Space is not a container but a property of the rope network. There is no space where there are no ropes.
Standard model: Space is a primitive background manifold.
SPACETIME
The rope network together with the timing of wave propagation. Curved spacetime = non-uniform rope tension $T(r)$. The metric tensor encodes the local T and μ of the rope network. General relativity equations $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$ describe how matter (rope knots) determines local rope tension.
See also: TENSION, GEODESIC
DISTANCE d [m]
The length of the rope connecting two points. Measured by counting how many rope wavelengths fit between them. The metric is $g_{ij} = \lambda_{ij}/\lambda_{ref}$ (ratio of local rope spacing to reference). In a non-uniform tension field, distances are curved.
SIMULTANEITY
Two events are simultaneous if they occur at the same phase of the rope wave. Relativity of simultaneity: different observers' ropes are in relative motion, so their wave phases don't agree. The rope gives a preferred local simultaneity (proper time) but no global absolute simultaneity.
LORENTZ FACTOR $\gamma = 1/\sqrt{(1-v^2/c^2)}$
In rope language: the compression ratio of rope waves in the direction of motion of a kink moving at speed v. A kink moving at $v \simeq c$ compresses its pilot wave in the forward direction (blue-shift) and stretches it aft (red-shift). Time dilation and length contraction follow from this wave geometry.
CLOCK
Any periodic process in the rope network. The fundamental clock is one oscillation of a kink's pilot wave. Time flows at the rate of these oscillations. Clocks slow down in a gravitational field because the rope tension is higher there, which changes μ and the wave speed.
PROPER TIME τ [s]

The time measured by a clock travelling with the object — the number of rope wave oscillations experienced by the kink on its worldline. Invariant under Lorentz boosts (it's a count of oscillations, not a rate).

WEYL SPINOR χ_L, χ_R **★NEW**

A one-component (single-strand) massless rope excitation. The right-mover χ_R satisfies $\bar{\sigma}^\mu \partial_\mu \chi_R = 0$; the left-mover χ_L satisfies $\sigma^\mu \partial_\mu \chi_L = 0$. Massless because no inter-strand coupling. Two Weyl spinors coupled by the mass term give the Dirac equation. In rope language: χ_R = strand-1 excitation; χ_L = strand-2.

See also: DIRAC SPINOR, DIRAC EQUATION, STRAND

DIRAC SPINOR $\psi = (\chi_R, \chi_L)^T$ **★NEW**

The four-component field arising from the two-strand rope. Left strand → χ_L; right strand → χ_R. The Dirac equation couples them with the mass term $m_e c^2/\hbar$. Pauli matrices arise from helix geometry: σ^1 from same-direction strand coupling; σ^2 from 90° helical phase lag (giving the imaginary i); σ^3 from strand tension asymmetry.

See also: DIRAC EQUATION, WEYL SPINOR, PAULI MATRICES

ZITTERBEWEGUNG ★ ★NEW

The rapid trembling motion of the electron kink between left and right strand modes at frequency $2m_e c^2/\hbar \approx 2.5\times10^{21}$ Hz. The particle zigzags because the mass coupling constantly converts left-movers to right-movers and back. Not directly observable (too fast) but implicit in the Dirac equation's α matrices.

See also: DIRAC EQUATION, MASS

5. Forces and Interactions

ELECTRIC FORCE

The force from a tension gradient in the rope network. A charged rope knot (winding $n \neq 0$) creates a radial tension gradient $\nabla T \sim 1/r^2$. The force on another knot is $F = \nabla T$ (tension gradient = Coulomb force). The electric field $E = -\nabla T/\varepsilon_0$ in rope language.

MAGNETIC FORCE

The force from the rotation of the tension gradient around a current. A moving rope kink (current) creates a circular tension gradient pattern. The magnetic force is $F = v \times B$ (where $B = \nabla \times T/c$). The derivation: $\nabla\cdot B = 0$ is a geometric theorem (tension gradient has no sources or sinks in vacuum).

GRAVITATIONAL FORCE

The force from a radial gradient in rope density or tension: $g(r) = -\nabla T/(\mu c^2)$. At large scales: the rope network's tension profile $T(r) \sim A r^{(2-\gamma)}$ reproduces observed rotation curves of 171 SPARC galaxies without dark matter halos. The rotation curve test: 152/171 galaxies have lower reduced χ^2 than NFW dark matter.

See also: RADIAL ACCELERATION RELATION, DARK ENERGY

PARITY VIOLATION ★ ★NEW

The weak force's preference for left-handed particles. In rope language: the helical rope is left-handed (chirality). The left-handed helix couples preferentially to χ_L (left-movers). The Standard Model's V−A coupling $J^\mu_{weak} = 2\bar{\chi}_L \sigma^\mu \chi_L$ arises from the helix chirality. Under parity (which reflects the helix) the coupling would go to χ_R — but the universe's helix is left-handed, so parity is violated.

Standard model: Parity violation is an experimental fact incorporated into the Standard Model via the V−A structure, but not geometrically explained.

See also: CHIRALITY, WEAK FORCE, WEINBERG ANGLE

WEAK FORCE ★ ★NEW
The force responsible for beta decay and flavour-changing processes. In rope language: massive two-strand rope kinks. W± bosons carry strand tension imbalance (charge ±e) and have mass M_W = 80.4 GeV from the spring term of the two-strand rope at electroweak scale. Z^0 is a neutral mix of EM and weak strand modes. The mixing angle is the Weinberg angle θ_W = θ_w.
See also: BETA DECAY, WEINBERG ANGLE, KLEIN-GORDON EQUATION
BETA DECAY ★ ★NEW
n → p + e^- + $\bar{\nu}$_e. In rope language: (1) a d quark sub-knot (n = −1/3) emits a W^- rope kink via its left-handed strand coupling; (2) the sub-knot reconfigures to u (n = +2/3), winding number change +1; (3) the W^- propagates ~0.0025 fm (range from M_W) and creates an electron kink (n = −1) and antineutrino. Winding number conserved: (−1/3) = (+2/3) + (−1). The V−A structure: only χ_L couples to the W^-, from rope chirality.
See also: WEAK FORCE, PARITY VIOLATION, CHARGE
WEINBERG ANGLE θ_W **★ ★NEW**
The mixing angle between the EM rope mode (B) and weak rope mode (W_3) in the two-strand helix. Physical mass eigenstates: photon = cos(θ_w)B + sin(θ_w)W_3 (massless); Z = −sin(θ_w)B + cos(θ_w)W_3 (massive). Measured at M_Z: sin²(θ_W) = 0.23122 ⇒ θ_w = 28.74°. From birefringence: θ_w ∼ 30°. Agreement to within 1.26° (4%), consistent with running of sin²(θ_W) from M_Z to low energies.
See also: WEINBERG-BIREFRINGENCE CONSISTENCY, CHIRALITY, WINDING ANGLE
6. Electromagnetic Fields
CHARGE AS HANDEDNESS / LINKING ▲UPDATEDv4 ★
Electric charge is the geometric handedness of the two strands — the glove-like mirror asymmetry distinguishing electron from positron — and this is one and the same integer as the linking (winding) number of the strands. Gaede's handedness ontology and the topological description (charge = linking number = first Chern class) are the SAME computed integer at two levels, verified to machine precision (electron/positron sum to zero handedness; |Lk| ≈ 1). Charge is therefore quantised in whole units because a fraction of a linking is not possible for closed strands.
Standard model: charge is a conserved quantum number attached to fields by fiat (Noether charge of a U(1) symmetry); its integer quantisation is imposed, not explained mechanically.
See also: WINDING ANGLE, LINKING NUMBER, SCREW CURRENT
SCREW CURRENT ★NEWv4
Electric current as the turning of the two-strand helix like a corkscrew or Archimedes screw: because a helix IS a screw, rotating it converts turning into axial transport with one motion, not two. This handles instant-on (torque transmits promptly as a fast torsional wave, not rigid lockstep), and the handedness (charge sign) sets the screw direction, hence the current direction. One turn pumps one unit of linking, realising charge continuity; power = torque × angular rate mirrors V × I.
Standard model: current is charge flow ρv; the "why" of the drift and its instant response are not given a mechanical picture.
See also: CHARGE AS HANDEDNESS / LINKING, CONTINUITY EQUATION
ELECTRIC FIELD E [V/m]
The tension gradient in the rope network: $E = -\nabla T/\varepsilon_0$. Coulomb's law follows from the 1/r² fall-off of the tension gradient around a point knot. The field exists at every point in the rope network where there is a tension gradient, not merely where there is charge.

MAGNETIC FIELD B [T]
The curl of the tension gradient: $B = \nabla \times A/c$ where A is the vector potential (rope displacement field). $\nabla \cdot B = 0$ is a geometric theorem (a curl has no divergence), not a separate physical law. The four Maxwell equations are all consequences of rope geometry.
VOLTAGE V [V]
The tension difference along a rope path: $V = -\int E \cdot dl = \Delta T/e$. A battery maintains a tension difference by chemical energy stored in rope knot configurations. Current flows when knots (electrons) move down the tension gradient.
CURRENT I [A]
The flow of rope kinks (electrons) along a conductor: $I = dQ/dt = e\, dn/dt$. In rope language: a tension wave propagating along the conductor rope bundle, carrying kinks. Ohm's law $V = IR$ arises from rope tension gradients driving kink flow against rope bundle resistance.
RESISTANCE R [Ω]
The opposition to kink flow through a conductor. In rope language: the energy dissipated when kinks scatter off rope knot configurations (phonons, impurities). Temperature dependence: more thermal rope fluctuations → more scattering → higher resistance.
CAPACITANCE C [F]
The ability to store tension energy in a separated rope configuration. A capacitor stores energy in the tension gradient between two plates (opposite winding knots). $E = CV^2/2$ = (tension energy stored in rope network between plates).
7. Special Relativity
SUB-NUCLEAR MESH / LORENTZ SCALE ★NEWv4
The requirement that the fundamental rope spacing a is far finer than atomic. The rope wave equation is exactly Lorentz-invariant in form (no detectable rest frame at the wave level), and the strand mechanics forbid the Galilean convective term that a moving material medium would have. A residual lattice-scale Lorentz violation of order $(ka)^2$ bounds the spacing: laboratory and astrophysical limits force $a < \sim 10^{-16}$ m — sub-nuclear, well above the Planck scale.
Consequence: the "ropes between atoms" image is a convenience, not the fundamental scale.
Standard model: Lorentz invariance is a postulate of special relativity; a fundamental discreteness scale, if any, is unspecified.
See also: ROPE NETWORK, PROTON STAR / CONVERGENCE
RELATIVISTIC MOMENTUM $p = \gamma mv$
The momentum of a rope kink moving at speed v. In rope language: the kink's pilot wave has a momentum $p = \hbar k$ where k is the wave vector. Relativistic momentum follows from the wave dispersion relation $\omega = c\sqrt{(k^2 + m^2c^2/\hbar^2)}$.
GEODESIC
The path of minimum tension along the rope network: the rope takes the path of least elastic deformation. In curved spacetime (non-uniform T): geodesics curve around massive objects. Free-falling objects follow geodesics because the rope tension gradient creates a force (gravity) that exactly curves their rope path.
8. General Relativity
SCHWARZSCHILD RADIUS $r_s = 2GM/c^2$
The radius at which the rope tension gradient becomes so steep that $c_{eff}(r) \to 0$ and outward propagation becomes impossible. Not where $T = 0$ (tension is large near the Schwarzschild radius) but where $\kappa(r) = 1 - r_s/r = 0$: the outward propagation capacity vanishes.
See also: OUTWARD PROPAGATION CAPACITY, EVENT HORIZON

OUTWARD PROPAGATION CAPACITY $\kappa(r) = 1 - r_s/r$
The capacity of a rope wave to propagate outward at radius r near a massive object. $\kappa = 1$ far from the mass (free propagation); $\kappa = 0$ at the Schwarzschild radius (all propagation inward). The event horizon is where $\kappa = 0$, NOT where tension is zero. Inside the event horizon: all rope wave directions are inward.

GRAVITATIONAL WAVE
A ripple in the rope tension field: a transverse oscillation of the rope network geometry propagating at c. In rope language: a quadrupolar distortion of the rope bundle density. Polarisations + and × correspond to the two quadrupolar oscillation modes of the rope network. The rope model predicts a possible scalar breathing mode not present in standard GR; LIGO data can test this.

GRAVITATIONAL REDSHIFT
The decrease in photon energy as it climbs out of a gravitational well. In rope language: the rope tension T(r) increases toward a massive object. Higher T means higher wave speed $c(r) = \sqrt{T/\mu}$, so outgoing photons gain wavelength (redshift) as they travel up the tension gradient. $\Delta\nu/\nu = GM/(rc^2)$.

SINGULARITY
In standard GR: a point of infinite density where the theory breaks down. In rope language: a region where all ropes converge and the concept of ‘between’ becomes ill-defined. The rope model does not resolve the singularity; it requires quantum rope mechanics at the Planck scale, not yet developed.

9. Black Holes

EVENT HORIZON ★
The surface where the outward propagation capacity $\kappa(r) = 0$. The rope tension at the horizon is ENORMOUS (not zero), entirely directed inward. A rope kink at the horizon cannot propagate outward because the tension gradient always exceeds the kinetic energy. Common misconception: ‘tension goes to zero at the horizon’ — this is incorrect. The horizon is a kinematic condition, not a mechanical one.
See also: OUTWARD PROPAGATION CAPACITY, SCHWARZSCHILD RADIUS

10. Chemistry and Atomic Structure

STANDING-WAVE SHELL / $2n^2$ ★NEWv4
Electron shells as standing torsion patterns on the ropes around the core. The number of independent patterns on a sphere up to level n is n^2 (spherical harmonics); the rope’s TWO strands supply a factor of 2, giving shell capacities $2n^2$ = 2, 8, 18, 32, 50 — the two-strand structure providing, with no extra postulate, the doubling that standard physics attributes to electron spin.
Standard model: $2n^2$ comes from n^2 orbitals times two spin states, spin being a separately postulated two-valued property.
See also: HYDROGEN (IRREDUCIBLE ATOM), STANDING WAVE, SPIN

HYDROGEN (IRREDUCIBLE ATOM) ★NEWv4
Hydrogen as the simplest possible atom for a structural reason: Z = 1 is the minimum non-zero winding (charge quantisation), so nothing simpler can exist; and a single winding has nothing to couple to, so hydrogen is the unique single-mode atom — matching its status as the only atom exactly solvable in standard quantum mechanics.
See also: STANDING-WAVE SHELL / $2n^2$, WINDING NUMBER

ATOMIC ORBITAL

A standing wave pattern of an electron kink guided by a rope mode around the nucleus. The rope mode shapes (s, p, d, f) are solutions to the rope mode equation (= Schrödinger equation). The electron is the kink; the orbital is the rope wave pattern guiding it. Orbitals are real standing waves, not probability distributions — the Born rule gives the probability of finding the kink at a given position within the orbital.

See also: PILOT WAVE, BORN RULE, ELECTRON

QUANTUM NUMBER n, l, m_l, m_s

Labels for rope wave modes. n = radial node count; l = angular node count (angular rope mode pattern); m_l = orientation of angular mode; m_s = strand choice ($\pm 1/2 \rightarrow$ which strand the kink occupies). Pauli exclusion: no two electrons in the same (n, l, m_l, m_s) because the exchange phase is -1.

COVALENT BOND

The sharing of electron rope modes between two atoms. When two hydrogen atoms approach, their 1s rope modes overlap. The bonding orbital is the symmetric combination (lower energy); antibonding is antisymmetric (higher energy). Bond energy = energy difference between shared mode and separate atomic modes. Covalent bonds are Level 2 dilute-regime forces ($N_{eff} \ll 1$), not Level 1 classical contact forces.

See also: TWO-LEVEL FORCE STRUCTURE, ATOMIC ORBITAL

IONIC BOND

Electron transfer between atoms, creating opposite-winding knot pairs at neighbouring sites. The resulting tension gradient between the opposite windings gives the ionic bonding energy (Coulomb energy in rope language).

VALENCE

The number of unfilled strand mode slots available for bonding. In rope language: the number of rope mode positions that can accommodate additional kinks. An atom with four unfilled slots (carbon) can form four bonds.

ELECTRONEGATIVITY

The tendency of an atom to pull shared rope mode density toward itself. In rope language: the steepness of the rope tension gradient at the nucleus determines how strongly it pulls the rope bundle of the bond toward itself.

HYBRIDISATION

The mixing of rope wave modes to form new mode shapes optimised for bonding. sp, sp^2, sp^3 hybrids are superpositions of s and p rope modes oriented toward bonding partners. Geometry of molecules (tetrahedral, trigonal planar, linear) follows from mode superposition geometry.

11. Thermodynamics

ENERGY E [J]

In rope language: the sum of elastic energy stored in rope waves and knots, kinetic energy of kink motion, and potential energy of tension gradients. Conservation of energy = conservation of rope elastic energy. Heat is incoherent rope wave energy (random phases, all directions); work is coherent rope wave energy (organised propagation).

ENTROPY S [J/K]

The number of distinct configurations of the rope network consistent with a given macroscopic energy. Maximum entropy = maximum rope configuration disorder (thermal equilibrium). The Second Law: rope configurations tend toward maximum disorder because there are exponentially more disordered configurations.

12. Cosmology and Gravity

DARK ENERGY
In rope language: the zero-point tension of the rope network. Every rope mode has zero-point energy $\hbar\omega/2$; summed over the entire network, this gives a vacuum energy that acts as a cosmological constant. The observed dark energy density $\Omega_\Lambda \approx 0.685$ may correspond to this rope vacuum energy, though the exact connection is not yet derived.
See also: ZERO-POINT ROPE FLUCTUATIONS, CASIMIR EFFECT
HUBBLE EXPANSION $H_0 = 67.4$ km/s/Mpc
The expansion of the rope network itself: ropes between distant galaxies are stretching. In rope language: T(t) decreases as the network expands, so c(t) changes cosmologically. The CMB redshift is a rope wavelength stretching with the network expansion.
COSMIC BIREFRINGENCE β ★
The rotation of CMB photon polarisation planes by $\beta = 0.342° \pm 0.094°$ (Eskilt & Komatsu 2022). In rope language: the Chern-Simons coupling $\chi = \sin(2\theta_w)\, n_{rope}\, r_h^2/c$ from the helical rope network preferentially rotates one circular polarisation vs the other as photons travel over cosmological distances. The angle β constrains the product $\sin(2\theta_w)\, n_{rope}\, r_h^2$.
See also: WEINBERG ANGLE, WINDING ANGLE, WEINBERG-BIREFRINGENCE CONSISTENCY
WEINBERG-BIREFRINGENCE CONSISTENCY ★ ★NEW
The discovery (Palmer & Claude 2026) that the rope winding angle θ_w measured from cosmic birefringence (EM sector, CMB photons, ~30°) and from the Weinberg angle (electroweak sector, M_Z scale, 28.74°) agree to within 1.26° (4%). The residual discrepancy is consistent with the running of $\sin^2(\theta_W)$ from M_Z down to CMB energies. Two independent measurements of the same geometric parameter from sectors 14 orders of magnitude apart in energy. Testable with LiteBIRD (2033).
See also: WEINBERG ANGLE, COSMIC BIREFRINGENCE, WINDING ANGLE
RADIAL ACCELERATION RELATION (RAR) ★ ★NEW
The empirical correlation $g_{obs} = g_{bar}/(1-\exp(-\sqrt{(g_{bar}/g†)}))$, $g† = 1.2\times10^{-10}$ m s^{-2} (McGaugh et al. 2016). In rope language: the rope model analytically predicts slope 1 at high g_{bar} (inner Newtonian regime where rope follows baryons) and slope 0.5 at low g_{bar} (outer regime where flat rotation curve V_{flat} = const gives $g_{rope} \sim 1/r$ while $g_{bar} \sim 1/r^2$). The rope and MOND produce identical RAR shapes because both produce flat rotation curves. Confirmed for 171 SPARC galaxies (rope beats NFW in 152/171).
See also: MOND ACCELERATION CONSTANT, BARYONIC TULLY-FISHER RELATION
MOND ACCELERATION CONSTANT $g† = 1.2\times10^{-10}$ m s^{-2} **★NEW**
The transition scale between Newtonian ($g_{obs} = g_{bar}$) and deep-modified ($g_{obs} = \sqrt{(g†\, g_{bar})}$) regimes. In rope language: g† is the centripetal acceleration at the rope tension transition radius r_0, where baryonic and rope contributions are equal. Universality of g† across galaxies reflects universality of the rope tension scale. Identified as $g† = V^4_{flat}/(G\, M_{bar}) = A^2/(G\, M_{bar})$.
See also: RADIAL ACCELERATION RELATION, BARYONIC TULLY-FISHER RELATION
BARYONIC TULLY-FISHER RELATION $M_{bar} \propto V^4_{flat}$ **★NEW**
The tight empirical correlation between galaxy baryon mass and flat rotation curve speed. In rope language: equivalent to $A^2 = g†\, G\, M_{bar}$ where A is the rope amplitude parameter. This is a consequence of the flat rotation curve and the MOND/rope transition: $g_{rope} = A/r$ while $g_{bar} = GM/r^2$, which gives $g_{rope} \sim \sqrt{g_{bar}}$ only if $A^2 = g†\, GM$. Confirmed for 171 SPARC galaxies.

See also: MOND ACCELERATION CONSTANT, RADIAL ACCELERATION RELATION

13. Quantum Mechanics

BORN RULE ★

The probability of finding a particle at position x equals $|\psi(x)|^2$. In rope language: the single-detector click RATE is benchmarked to scale as $|\psi|^2$ under threshold (nucleation) dynamics in the weak-field regime (QB-007); the interpretation of this scaling as the Born rule is CONDITIONAL on the threshold model being the correct physical description of detection, which is itself part of the conjecture under test — not a completed derivation.

See also: PILOT WAVE, WAVEFUNCTION

QUANTUM TUNNELLING

The passage of a rope kink through a tension barrier higher than its kinetic energy. In rope language: the pilot wave is not confined by the barrier (waves penetrate barriers as evanescent modes), so the kink has a non-zero probability of being found on the other side. The tunnelling amplitude is the evanescent rope wave amplitude through the barrier.

WAVEFUNCTION $\psi(x) = \langle x|\psi\rangle$

The amplitude of the rope pilot wave at position x. In rope language: $\psi(x)$ is the complex displacement of the rope at x, encoding both the amplitude and phase of the local rope oscillation. The Schrödinger equation is the rope mode equation for the pilot wave.

See also: PILOT WAVE, BORN RULE

ENTANGLEMENT ★NEW

The registered BOUNDARY of the rope programme, not a solved item. A configuration-counting rope model provably cannot reproduce Bell/CHSH violation (QB-003 Failed, QB-005 negative). The single-particle statistics ARE mapped — indivisibility (from integer topology), Born-rate scaling, single-photon anticorrelation (QB-007–009) — and the residual gap is localized precisely to configuration-space guidance (QB-010): the identical race gives CHSH S = 1.42 in physical space vs 2.83 only when the configuration-space object is imported by hand. Retired as an ambition, preserved as an honest limit; a future non-classical rope structure is not claimed impossible, and none is claimed to exist.

See also: CHSH BOUNDARY, MEASUREMENT DECOMPOSITION, PILOT WAVE, BORN RULE

CHSH BOUNDARY ★NEW

The quantitative wall between classical and quantum correlation. Local models are bounded by $S \le 2$; quantum mechanics and experiment reach $S \approx 2.83$ (the Tsirel'son bound). The rope model reproduces the correlation's FUNCTIONAL FORM only when granted the nonlocal conditional (see ENTANGLEMENT); its own physical-space mechanics give $S \approx 1.42$. The boundary is mapped to high precision, not crossed.

See also: ENTANGLEMENT, GAMMA IDENTIFICATION, MEASUREMENT DECOMPOSITION

MEASUREMENT DECOMPOSITION ★NEW

The decomposition of the measurement problem into separable pieces (QB-007). INDIVISIBILITY — a detector fires all-or-nothing — is DERIVED from integer topology (you cannot tie half a knot). The single-site click RATE scales as $|\psi|^2$ under threshold dynamics (benchmarked, conditional on the threshold model). ANTICORRELATION — that only one detector fires — is the irreducible core: a classical threshold model is pinned at $g^2(0) \ge 1$ versus the measured ≈ 0.18, a registered negative repaired only by the conservation toy's imported instantaneous budget (QB-009).

See also: ENTANGLEMENT, BORN RULE, CHSH BOUNDARY

GAMMA IDENTIFICATION (γ = 1) ★NEW

The detector-coupling angle that decides whether the rope model reproduces the quantum cosine. Fixed to $\gamma = 1$ by three independent constraints (QB-011): STRUCTURE — an energy detector squares the spinor half-angle amplitude, $|\cos(\theta/2)|^2 = (1+\cos\theta)/2$, projecting onto the S^2 base of the Hopf bundle, fibre-blind; CONSISTENCY — setting-relabeling antisymmetry forces γ to an odd integer, excluding $\gamma = ½$ a priori; DYNAMICS — the QB-007 threshold exponent $p \approx 2$ gives $\gamma = p/2 \approx 1$, excluding $\gamma = 3$. Note: distinct from the Lorentz factor γ and the twist–stretch locking constant $\gamma = 1/\sin^2\theta$, which share the symbol. Conditional on the threshold model; does not itself cross the CHSH wall.

See also: CHSH BOUNDARY, ENTANGLEMENT, MEASUREMENT DECOMPOSITION

FERMIONIC STATISTICS ★ ★NEW

The property that no two identical fermions can occupy the same quantum state (Pauli exclusion). In rope language: the two-strand helix creates a half-twist (π phase) when two topological knots are exchanged. Exchange phase $e^{i\pi} = -1$ means antisymmetric wavefunction $\psi(x_A, x_B) = -\psi(x_B, x_A)$. For two identical particles at the same position: $\psi(x, x) = -\psi(x, x) = 0$. The spin-statistics theorem is thus a consequence of rope topology.

See also: PAULI EXCLUSION, SPIN, TWO-LEVEL FORCE STRUCTURE

PAULI EXCLUSION ★ ★NEW

The impossibility of two electrons occupying the same quantum state (n, l, m_l, m_s). In rope language: derived from fermionic statistics. The exchange of two electron knots via the two-strand helix creates a phase −1. For identical states (same position and quantum numbers): $\psi = -1 \times \psi \Rightarrow \psi = 0$. Not a postulate — a topological consequence.

See also: FERMIONIC STATISTICS, SPIN

DIRAC EQUATION $(i\hbar\gamma^\mu\partial_\mu - mc)\psi = 0$ **★NEW**

The relativistic wave equation for spin-1/2 particles. In rope language: the coupled wave equations for the two strands of the rope, with mass term from the inter-strand coupling: $(\partial/\partial t + c\partial/\partial z)\chi_L = -(mc^2/\hbar)\chi_R$ and vice versa. Pauli matrices arise from helix geometry. Predicts: antimatter (opposite-winding solutions), spin-1/2 (half-twist exchange), parity violation (chiral coupling), Zitterbewegung.

See also: DIRAC SPINOR, WEYL SPINOR, PAULI MATRICES, ANTIMATTER

14. Nuclear Physics

STRONG FORCE AS BUNDLE CONTACT ★NEWv4 ★

The strong nuclear force identified as the rope-bundle contact force — the same “you can’t be in my space” that makes solid matter solid — operating at nuclear scale ($\sim 10^{-15}$ m) rather than atomic scale, hence $\sim 10^5\times$ denser and correspondingly stronger. Explains the strong force’s peculiarities as features of contact: short range with a sharp cutoff (bundles either overlap or don’t), charge independence (contact ignores the twist that sets charge), and a hard repulsive core plus attractive well (bundles jamming vs. just touching). Qualitative identification; the quantitative Yukawa potential is an open problem.

Standard model: the strong force is gluon exchange (QCD); precise but not mechanically picturable — gluons carry their own charge and the force does not fall off with distance.

See also: ROPE BUNDLE, QCD STRING, NUCLEON KNOT

NUCLEON KNOT ★NEWv4

A proton or neutron as a complete three-sub-knot cluster (uud, udd). The neutron has zero NET winding (neutral) but is a full knot with full mass — so the mass-bearing count is the nucleon-

knot number A (protons + neutrons), NOT the net winding (charge) Z. This is why atomic weight tracks A, not charge.
See also: QUARK SUB-KNOT, NUCLEAR BINDING (MODE OVERLAP), ATOMIC MASS

NUCLEAR BINDING (MODE OVERLAP) ★NEWv4

Nuclear binding energy identified as rope mode-overlap energy — the nuclear analogue of a chemical bond — largest where knot packing is optimal, hence the binding-per-nucleon peak at iron-56. The nucleus weighs less than its free parts by this binding energy (the mass deficit). Mechanism proposed and consistent; the binding curve is not yet computed from rope parameters (binding values are read from experiment).
See also: STRONG FORCE AS BUNDLE CONTACT, ATOMIC MASS

ATOMIC MASS (FROM NUCLEON KNOTS) ★NEWv4

An atom’s mass as the sum of its nucleon-knot masses minus the nuclear binding (mass deficit). Given measured binding energies this reproduces real atomic masses to better than 0.1% across the periodic table; the naive “mass = charge/winding” fails by up to ~2.6× because it is blind to neutrons. Reproduction-given-binding plus a mechanism, not yet a from-scratch prediction.
See also: NUCLEON KNOT, NUCLEAR BINDING (MODE OVERLAP)

QCD STRING ★ ★NEW

The rope bundle between two quark sub-knots, providing the strong force at quark scale. The QCD string tension k_QCD = T_nuclear = u_nucleon × λ_c_pion^2 ≈ 1.17×10^5 N (observed: 1.46×10^5 N, within 20%). A Level 1 direct rope bundle contact force. The string energy grows linearly with quark separation: E = k_QCD × r, which is confinement.
k_QCD = (m_p c^2 / V_nucleon) x (hbar / m_pi c)^2 = 1.17e5 N
See also: CONFINEMENT, T = u×λ_c^2, ROPE BUNDLE

CONFINEMENT ★ ★NEW

The impossibility of isolating a single quark. In rope language: quark sub-knots are connected by QCD string rope bundles. Pulling a quark out extends the bundle, storing energy at rate k_QCD ~ 10^5 N per metre, until sufficient energy for pair creation makes it cheaper to create a new quark-antiquark pair than to extend the string further. The quarks are permanently confined because the QCD string energy grows without bound.
See also: QCD STRING, PAIR CREATION/ANNIHILATION

NUCLEAR MATTER DENSITY n_nuclear = 1.38×10^{44} m^{-3} **★NEW**

The density at which pion-scale rope bundles first satisfy the contact condition N_eff = 1. Predicted (no free parameters): n_nuclear = (m_π c/ℏ)^3, giving bounds 7.6×10^{43} to 3.2×10^{44} m^{-3} bracketing the observed 1.38×10^{44} m^{-3}. Physical meaning: nuclear matter packs nucleons one per pion Compton sphere. Nuclear saturation density is the pion-scale rope bundle contact threshold.
n_nuclear = (m_pi c / hbar)^3 [contact threshold at pion coherence length]
See also: CONTACT CONDITION, PION, NUCLEAR SATURATION

PION π, m_π = 135 MeV **★NEW**

The lightest massive rope kink on the quark bundle system. Its mass m_π = 135 MeV gives the spring term k_s = m_π^{2}c^2/ℏ^2 in the rope wave equation (Klein-Gordon), making the nuclear force short-range with λ_π = ℏ/(m_πc) = 1.46 fm. The pion is to the nuclear force what the photon is to the EM force — but massive (from the spring term).
See also: KLEIN-GORDON EQUATION, YUKAWA POTENTIAL, NUCLEAR MATTER DENSITY

NUCLEAR SATURATION ★ ★NEW

The phenomenon that nuclear matter reaches a fixed density (~0.17 nucleons/fm^3) regardless of nucleus size. In rope language: this is the pion-scale rope bundle contact threshold density. Below this density, the nuclear force is Yukawa (residual); at this density, the rope bundles first become collectively solid (contact regime). Nuclear matter cannot be compressed further without exceeding the hard core (Level 1 direct contact at proton Compton scale).

See also: NUCLEAR MATTER DENSITY, CONTACT CONDITION, PION

YUKAWA POTENTIAL $V(r) = -g^2 \exp(-r/\lambda_\pi)/(4\pi r)$ ★NEW

The nuclear potential from one-pion exchange. In rope language: the static Green's function of the massive rope wave equation (Klein-Gordon with spring term $k_s = m_\pi^2 c^2/\hbar^2$). The pion propagates from nucleon A to nucleon B, mediating a Yukawa interaction. A Level 2 RESIDUAL force from quark bundle fluctuations, analogous to van der Waals being a residual EM force. Range λ_π = 1.46 fm set by pion Compton wavelength.

See also: KLEIN-GORDON EQUATION, PION, RESIDUAL FORCE

NUCLEAR BINDING ENERGY ~8.8 MeV/nucleon (Fe) ★NEW

The energy released when nucleons form a nucleus. A Level 2 RESIDUAL force from fluctuations in the quark rope bundles of neighbouring nucleons — exactly analogous to van der Waals being a residual force from electron bundle fluctuations. The binding energy is ~8.8 MeV, comparable to but less than the QCD string energy at 1.5 fm (~1367 MeV), confirming the residual nature.

See also: RESIDUAL FORCE, YUKAWA POTENTIAL, TWO-LEVEL FORCE STRUCTURE

15. Rope Bundle Theory

ROPE DENSITY $n_{rope} = (2\pi/3) n_{local}^2 \lambda_c^3$ [m^{-3}] ★NEW

The effective number of rope pairs per unit volume at coherence scale λ_c. Derived by counting rope pairs within one coherence volume: n_{local} atoms in volume λ_c^3, each with $n_{local} \times (4\pi/3)\lambda_c^3$ partners, divided by 2 to avoid double-counting and by λ_c^3 to get volume density. The contact condition $N_{eff} = n_{rope} \times \lambda_c^3 = (2\pi/3) n_{local}^2 \lambda_c^6 \gg 1$ determines whether classical bundle contact theory applies.

n_rope = (2*pi/3) * n_local^2 * lambda_c^3

CONTACT FORCE $E = (1/8) n^2 T \lambda_c^3 V$ ★NEW

The Level 1 force from dense rope bundles ($N_{eff} \gg 1$). The energy of interaction between two overlapping bundles of density n_1, n_2 over volume $V_{overlap}$: $E = (1/8)\, n_1 n_2\, T\, \lambda_c^3\, V_{overlap}$. Applications: QCD string tension ($T = T_{nuclear}$, n = quark density), macroscopic solidity ($T = T_{atomic}$, n = atomic density).

See also: ALPHA COEFFICIENT, $T = u \times \lambda_c^2$

RESIDUAL FORCE ★ ★NEW

A Level 2 force arising from quantum fluctuations in Level 1 rope bundles. Requires quantum rope perturbation theory; cannot be computed from classical bundle contact formula. The structural analogy: Coulomb force (Level 1) → van der Waals (Level 2); QCD string (Level 1) → nuclear Yukawa (Level 2). Residual forces are generically weaker than their parent Level 1 forces and have different distance dependence.

See also: TWO-LEVEL FORCE STRUCTURE, VAN DER WAALS FORCE, NUCLEAR BINDING ENERGY

ALPHA COEFFICIENT $\alpha = 1/8$ ★ ★NEW

The geometric prefactor in the rope bundle contact force formula. Derived analytically (not fitted) as the product of three averages: time average (1/2, from oscillatory wave energy), angular average (1/3, from isotropic rope directions), position average (3/4, from kink position distribution within mode). Product: $(1/2)(1/3)(3/4) = 1/8$. This is the fundamental rope bundle geometric factor.

TWO-LEVEL FORCE STRUCTURE ★ ★NEW

The discovery that physical forces divide into exactly two categories: Level 1 (direct rope bundle contact forces, classical, $N_{eff} \gg 1$) and Level 2 (residual forces from quantum fluctuations in Level 1 bundles). QCD string, hard core, and macroscopic solidity are Level 1. Nuclear Yukawa force and van der Waals force are Level 2. The hierarchy: Level 1 is stronger; Level 2 requires quantum rope perturbation theory. Identical in structure to the hierarchy Coulomb / van der Waals in electrodynamics.

See also: CONTACT FORCE, RESIDUAL FORCE, NUCLEAR BINDING ENERGY, VAN DER WAALS FORCE

16. Quantum Rope Field

QUANTUM ROPE MECHANICS ★ ★NEW

The canonical quantisation of the rope wave field: promoting the classical displacement $u(z,t)$ to a quantum field operator $\hat{u}(z,t)$ with commutation relation $[\hat{u}(z,t), \hat{\pi}(z',t)] = i\hbar\delta(z-z')$. The Hamiltonian $H = \Sigma\hbar\omega_k(a_k^\dagger a_k + 1/2)$ is identical in structure to QED. Gives: zero-point rope fluctuations, Casimir effect, van der Waals forces, Hawking radiation.

See also: ZERO-POINT ROPE FLUCTUATIONS, CASIMIR EFFECT, VAN DER WAALS FORCE

ZERO-POINT ROPE FLUCTUATIONS ★ ★NEW

The non-zero vacuum expectation $\langle 0|\hat{u}^2|0\rangle \neq 0$ in the quantised rope field. Every rope mode has zero-point energy $\hbar\omega/2$. Physical consequences: Casimir effect (boundary restriction of modes), van der Waals (correlated fluctuations between atoms), Hawking radiation (mode splitting at event horizon), cosmological constant (sum of all mode zero-point energies). The vacuum is not empty — it is full of zero-point rope oscillations.

See also: CASIMIR EFFECT, VAN DER WAALS FORCE, DARK ENERGY

CASIMIR EFFECT ★ ★NEW

The attraction between parallel conducting plates from zero-point rope fluctuations. The plates restrict which rope modes can exist between them (wavelengths > 2d are excluded). The restricted zero-point energy is lower, pulling the plates together. $E = -\pi^2\hbar c/(720d^3)$ per unit area. At d = 100 nm: P = 13 Pa. Measured to 1% precision (Lamoreaux 1997). The rope model gives the exact result because it contains QED as a special case.

P_Casimir = -pi^2 hbar c / (240 d^4) [confirmed at d=100nm: P=13 Pa]

See also: ZERO-POINT ROPE FLUCTUATIONS, QUANTUM ROPE MECHANICS

VAN DER WAALS FORCE C_6/r^6 **★ ★NEW**

The Level 2 residual force between neutral atoms from correlated electron rope bundle fluctuations. The electron kink at a definite position (Bohmian picture) creates an instantaneous tension dipole; second-order quantum rope perturbation theory gives $C_6 = (3/4)\alpha_{pol}^2 I_H$ (London 1930 formula). For hydrogen: $C_6 = 7.59$ a.u. (observed 6.50, 17% from single-frequency approximation; exact Casimir-Polder gives 6.50 exactly). A Level 2 residual force requiring quantum rope mechanics.

C6(H) = (3/4) * alpha_pol^2 * I_H = 7.59 a.u. [obs: 6.50 a.u.; 17% from approx]

See also: LONDON FORMULA, ZERO-POINT ROPE FLUCTUATIONS, RESIDUAL FORCE

KLEIN-GORDON EQUATION ★ ★NEW

The wave equation for a massive rope field with spring term k_s: $\mu\partial^2u/\partial t^2 = T\nabla^2u - k_s u$. Dispersion: $\omega^2 = c^2(k^2 + m^2c^2/\hbar^2)$ with effective mass $m = \hbar\sqrt{(k_s/T)}/c$. Static Green's function:

the Yukawa propagator $G(r) = \exp(-r/\lambda)/(4\pi r)$. The pion satisfies this equation with $m = m_\pi$ = 135 MeV; the W and Z bosons satisfy it with m = 80.4 and 91.2 GeV respectively.
See also: PION, YUKAWA POTENTIAL, WEAK FORCE
LONDON FORMULA $C_6 = (3/4)\alpha_{pol}^2\hbar\omega_0$ **★NEW**
The van der Waals coefficient from second-order quantum rope perturbation theory. In rope language: α_{pol} is the rope mode's response to external tension gradient; $\hbar\omega_0 = I_H$ is the characteristic rope mode frequency. The prefactor 3/4 comes from the dipole-dipole angular average (different from the bundle contact $\alpha = 1/8$). The London formula is identical to rope second-order perturbation theory because the rope model contains EM as a special case.
See also: VAN DER WAALS FORCE, QUANTUM ROPE MECHANICS
END OF GLOSSARY — VERSION 4
137 terms · 16 categories · Version 4 (quantum-measurement arc addendum: ENTANGLEMENT, CHSH BOUNDARY, MEASUREMENT DECOMPOSITION, GAMMA IDENTIFICATION)
Terms marked ★ are those where the rope model provides a genuinely new physical explanation. Terms marked ★NEW were added in v3. Terms marked ▲UPDATED have expanded definitions from v2.
Palmer, M. & Claude 2026. The Rope Hypothesis: Scientific Glossary v4. NTT DATA / Anthropic working document.
Additional Terms: Extensibility, Calibration, Gravity, Quantum Boundary
Strand extensibility (k). The strand's resistance to being stretched along its length -- the fourth micro-primitive, alongside tension T_0, inertia μ, and torsional stiffness λ. Required for matter to exist: pure tension (k = 0) makes the network nonlinearly unstable. Generates the repulsive core with coefficient $c_4 = (k - T_0)/8$.
Volume-conservation postulate (P-VOL). Strands are volume-conserving filaments: compressing one lengthwise thickens it ($\text{width}^2 = w_0^2/(1+\varepsilon)$). Adopted 10 July 2026 as the minimal, zero-parameter premise that stops longitudinal relaxation from dissolving the repulsive core.
Vacuum tension density ($\Sigma = T_0 \cdot n_L$). The total rest tension the network carries per unit volume of empty space -- the single constant converting laboratory field strengths into strand strain ($g = E\sqrt{(\varepsilon_0/\Sigma)}$). PVLAS vacuum-birefringence limits force $\Sigma \gtrsim 10^{28}$ J/m³.
Dark longitudinal channel. Tension waves along strands, travelling faster than light (stability forces $k > T_0$). Gapless in principle (displacement is gauge), yet unable to carry signals at linear order because it decouples exactly from light and matter. A falsifiable prediction, and the candidate substrate for nonlocal quantum correlations (necessary, far from sufficient).
Twist–stretch locking ($\gamma = 1/\sin^2\theta$). Two-strand helix geometry ties stretching to untwisting, with the exact factor γ derived from the helix angle. Stiffens the longitudinal sector; famously does NOT gap it (an earlier gap mechanism was refuted and retracted).
Effective core stiffness ($k_{eff} = kK_c/(k+K_c)$). The repulsive core's surviving stiffness after strain is allowed to relax jointly into stretch and coverage channels; L-independent, so the core survives with the mild condition $k_{eff} > T_0$.
Zero-point theorem. A uniform network of any tension density is gravitationally flat: light and matter respond only to local network properties, so only deviations gravitate. Resolves the static half of the vacuum-energy problem for any value of Σ.
Network rigidity ($c^4/4\pi G$). What Newton's constant inverse-measures in the elastic-gravity picture: $\approx 9.6\times10^{42}$ N. Deriving G is provably equivalent to deriving the particle-mass hierarchy.

Quantum equilibrium. In pilot-wave theories, the requirement that hidden variables be Born-distributed. In the rope corpus this is noted as a CONDITION such theories assume, not a result the rope model derives; the model does not reproduce Bell/CHSH violation in its present form (see ENTANGLEMENT).

Additional Terms (Consolidation)

Strand extensibility (k). The fourth micro-primitive: how strongly a strand resists being stretched, beyond its rest tension T_0. Required for stable matter -- pure tension is nonlinearly unstable. Generates the repulsive core.

Extensibility core ($c_4 = (k-T_0)/8$). The super-quadratic term in the network's energy, produced by finite extensibility. Its positive sign is forced by stability. Sets bond lengths, nucleon spacing, ferromagnetic magnitude, and the strong-force range.

P-VOL (volume-conservation postulate). Strands are volume-conserving filaments: compressing thickens them, stretching thins them ($\text{width}^2 = w_0^2/(1+\varepsilon)$). Adopted (zero new parameters) to close the longitudinal-relaxation channel; protects the core.

Vacuum tension density ($\Sigma = T_0 \cdot n\_L$). The tension energy per unit volume carried by the vacuum's rope network. The one constant in the field–strain calibration $g = E\sqrt{(\varepsilon_0/\Sigma)}$. PVLAS bounds require $\Sigma \gtrsim 10^{28}$ J/m³.

Dark longitudinal channel. Tension waves along strands travel faster than light ($c\_L > c$ since $k > T_0$), are gapless in principle (longitudinal displacement is gauge), yet decouple from matter and light at linear order -- present but unable to signal.

Twist–stretch locking ($\gamma = 1/\sin^2\theta$). Helix geometry ties rope stretch to twist change; γ is the exact locking factor at helix angle θ. Derived; it stiffens (does not gap) the longitudinal sector.

Effective stiffness ($k\_eff = k \cdot K\_c/(k+K\_c)$). The local stiffness that survives when longitudinal relaxation is priced by the coverage ceiling (via P-VOL). Core condition: $k\_eff > T_0$.

Zero-point theorem (gravity). A uniform network of any tension density is gravitationally flat: metric perturbations depend only on deviations. Rest tension does not gravitate; only conditioning does.

Network rigidity ($c^4/4\pi G$). What Newton's constant inverse-measures in the elastic-gravity picture: $\sim 9.6\times10^{42}$ N. Deriving G reduces exactly to the particle-mass hierarchy.

Quantum equilibrium (fifth condition). For any nonlocal substrate to yield Bell correlations without signalling, its hidden variables must be Born-distributed. Stated here as an imported condition of pilot-wave accounts, NOT as something the rope model achieves — the configuration-counting form provably cannot cross the CHSH bound (see ENTANGLEMENT, CHSH BOUNDARY).